

How to write a report

Bioanalysis (FA-BA319/FA-CPS337)

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Based on: Linda McPhee, Linda Silvertand



Report

FA-BA319 and FA-CPS337 course guide

All students write an **individual report** about the developed method and its validation. The introduction and materials & methods may be a team effort, although every student needs to write **individually** the **results & discussion** section (containing the conclusions).

Instructions: see BB

Rubric: see BB

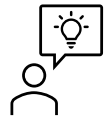
Deadline: last day of the course (**June 28th**) at 17:00

A: Report: Content						
1 Introduction (context and importance of project)						
2 Method (development and final method explained)						
3 Validation (well explained and sound calculations; other results)						
4 Conclusion (clear conclusion truly based on the results)						
5 Discussion (results in perspective; future plans)						
B: Report: Style						
1 Figures						
2 Connections and coherence						
3 Language						
4 Target group						
5 Length						

Objectives:



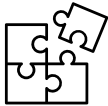
Discuss



Brainstorm



Draft



Organize



Feedback & improve



Apply!

Helpful, interactive, efficient, and...fun!

Work in process:

➤ *Your feedback and input is welcome!*

Structure:



Individual



**Project
Team**



Peers

Content:

Session 1 (May 27th):

1

- General
- Structure
- **Draft (*poster*)**
- Introduction: *context and importance*

Session 2 (June 10th):

2

- Material & Methods
- Results & Discussion
- **Figures**
- Conclusion
- Abstract



Outline
Introduction (literature review)
At least 1 result



How's your report writing?



*What does it make
easy/easier?*



*What difficulties do you
experience?*

A good report is...



Legible:

- ☐ Avoid spelling or grammatical errors
- ☐ Don't use unnecessarily difficult words.
- ☐ Use short sentences.
- ☐ Abbreviations: first time full out.

Concise:

Longer isn't necessarily better...

- ☐ Relevant information only.
- ☐ Keep it simple and direct.
- ☐ Avoid redundancy.

Complete:

- ☐ Transparent.
- ☐ Reproducible.

Language



Tense:

- Introduction and Discussion/Conclusion → *present tense*
- Materials & methods and Results → *past tense*
- Do not use future tense

Scientific:

- **Objective:** avoid superlatives (i.e., *groundbreaking*) and hyperbolic language (i.e., *extremely, amazing* or *enormous*).
- **Impersonal:** use first person sparingly
- **Formal:** no metaphors/slang
- **Accurate:** avoid indefinite words (i.e., *frequently* or *occasionally*) and list exact specifications.
- **Unambiguous:** be specific; repeat the subject instead of “this” or “that”.

Don't rely solely on your spell checker!

Ode to my spell checker (author unknown):

Eye halve a spelling checker
It came with my pea sea.
It plainly marks four my revue
miss steaks eye kin knot sea.

Eye strike a quay and type a word
and weight for it to say
Weather eye yam wrong oar write.
It shows me strait a weigh

As soon as a mist ache is maid.
It nose bee fore two long
and eye can put the error rite.
Its rare lea ever wrong.

Eye have run this poem threw it,
I am shore your pleased to no.
Its letter perfect awl the way.
My checker told me sew

Title page



- Title needs to be concise:
 - Less words is better
 - Describes the content accurately
 - Specific
- Name and student number
- Date

Title page



Good?

“Ethyl glucuronide”

Better?

“Bioanalysis of
Ethyl glucuronide in
urine”

The process



Content



Outline



Draft



Report



Coherent story

Today

What?



Content

Where?



Outline

How?



Draft

Set the context

Begin with the end in mind!

What is the bigger picture of your topic?

Where in the whole process are you?

Why is your research relevant?

- Formulate a clear and **SMART** goal.

Who is your reader?

- What is his background? What additional information is needed? What information can you skip?
- If you were an outsider to this research, what would you want to know?



Lay-out

Divide the report in sections and stick to it!



- Title page
- Content
- Abstract
- Introduction
- **M**aterials and Methods
- **R**esults
- **D**iscussion
- Conclusion
- Recommendations
- References
- Appendices (if applicable)

➤ Subdividing the sections (M&M, Results) into paragraphs often makes the report more legible.

Lay-out



Divide the report in sections and stick to it!

- Do **not** add new Methods in the Result section
- Do **not** put parts of the Introduction into the Discussion
- Use **page numbers**, also in the content sections
- Also number the pages of the appendices

Lay-out

Divide the report in sections and stick to it!



before the clear solution was transferred into glass vials. A volume of 10 μ l was injected into the LC–MS system.

2.6. Method validation

Validation was based on the FDA guidelines for Bioanalytical Method Validation [26].

2.6.1. Linearity

Ten calibration standards were prepared by serial dilution in concentrations of 0; 5; 10; 20; 50; 100; 200; 400; 700 and 1000 ng/ml above the endogenous concentration of HNP-1, -2 and -3, quantified as 7.4, 41.3 and 5.8 ng/ml, respectively. All standards were analyzed in each run, calculating target peptide: IS ratios of the peak area for each concentration level. Standard curves were constructed by least-squares linear regression analysis using a weighting factor of $1/x^2$ (with x as the concentration in ng/ml).

2.6.2. Precision and accuracy

Precisions and accuracies of the method for plasma and serum samples were determined by analysis of samples with endogenous concentrations and QC samples spiked at three different concentration levels in three separate runs ($n = 18$ at each level). Furthermore, one run contained plasma and serum spiked above the ULQ level at 5000 ng/ml, analyzed after 5-fold dilution with the same human plasma batch as used for the calibration standards to validate dilution of the samples. The accuracies of the recovered analytes were calculated as $[(\text{overall measured concentration} - \text{initial concentration}) / \text{added concentration}] \times 100\%$. Intra- and inter-assay precisions were expressed as relative standard deviations (RSD).

2.6.3. Selectivity

Human serum and plasma samples from six different sources were spiked at concentrations of 100 ng/ml and analyzed in triplicate. The plasma samples consisted of samples with EDTA, lithium heparine or citrate-phosphate-dextrose as anti-coagulant. For all sample batches, endogenous concentrations were mea-

were assessed after six months storage at -80°C . Deviations were calculated by comparing MS response ratios to freshly prepared samples at identical concentrations.

All stability tests were performed in triplicate and at three different concentration levels (LQC, MQC and HQC). The analytes were considered stable when 85–115% of the initial concentration was found.

2.7. Analysis of serum samples from intensive care patients

HNP-1, -2 and -3 concentrations were quantified in 24 serum samples obtained from 11 different patients at the intensive care unit. One serum sample of each patient was collected at 6.00 p.m., while additional samples of some patients were drawn on different days. Samples with concentrations beyond the ULQ of the assay were re-analyzed after 2- or 5-fold dilution of the sample with plasma from the same batch as used for preparation of the calibration standards. QC samples of all concentration levels were measured along with the patient samples to monitor the analytical performance.

3. Results and discussion

3.1. Method development

3.1.1. Sample pre-treatment

The protein precipitation procedure offered rapid sample preparation, facilitating high sample throughput. It furthermore provided better recoveries, especially for HNP-2, than solid-phase extraction (SPE) on different types of reversed-phase cartridges, although SPE is most commonly applied for sample clean-up of peptides from complex matrices [27]. Acidification of the organic solvent was required to prevent co-precipitation of the analytes. Addition of 200 μ l instead of 250 μ l acetonitrile resulted in substantially clearer extracts. Furthermore, evaporation of the supernatant appeared beneficial as interferences which were insoluble in the reconstitution solvent could be removed after reconstitution and subsequent centrifugation. However, as no solid precipitate was

Lay-out



1. van den Broek et al. / J. Chromatogr. B 878 (2018) 1085–1092 1087

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2.6. Method validation

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2.6.1. Linearity

Ten calibration standards were prepared by serial dilution in concentrations of 0.5; 10; 20; 50; 100; 200; 400; 700 and 1000 ng/ml above the endogenous concentration of HNF-1_α 2 and 3, quantified as 7.4, 41.3 and 5.8 ng/ml, respectively. All standards were analyzed in each run, calculating target peptide. Ratios of the peak area for each concentration level. Standard curves were constructed by least-square linear regression analysis using a weighting factor of 1/x² (with x as the concentration in ng/ml).

2.6.2. Precision and accuracy

Precision and accuracy of the method for plasma and serum samples were determined by analysis of samples with endogenous concentrations and QC samples spiked at three different concentration levels in three separate runs (n = 18 at each level). Furthermore, one run contained plasma and serum spiked above the ULQ level at 2000 ng/ml, analyzed after 5-fold dilution with the same human plasma batch as used for the calibration standards to validate dilution of the samples. The accuracies of the recovered analytes were calculated as [(overall measured concentration – initial concentration)/added concentration] × 100%. Intra- and inter-assay precisions were expressed as relative standard deviations (RSD).

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1. Introduction

2. Experimental

2.1 Chemicals and reagents

2.2 Instruments

2.3 LC-MS/MS conditions

2.4 Preparation of standards and quality control samples

2.5 Sample preparation

2.6 Method validation

2.6.1 Linearity

2.6.2 Precision and accuracy

2.6.3 Selectivity

2.6.4 Recovery and ion suppression

2.6.5 Stability

2.7 Analysis of serum samples from intensive care patients

3. Results and discussion

3.1 Method development

3.1.1 Sample pre-treatment

3.1.2 LC-MS/MS analysis

3.2 Method validation

3.2.1 Linearity

3.2.2 Accuracy and precision

3.2.3 Selectivity

3.2.4 Recovery and ion suppression

3.2.5 Stability

3.3. Internal standard and matrix selection

3.4 Analysis of serum samples from intensive care patients

4. Conclusion

Lay-out - exercise

Divide the report in sections and stick to it!

Individually:

- Create a **Table of Content** with *all sections*
- List the different **(sub-)paragraphs** in the Material & Methods and Results & Discussion sections. Include paragraphs for:
 - Expected experiments (based on ideal situation)
 - What you've already have done (and will report)
- Look at other research articles for examples (your own or on BB)
- Add notes and ideas to all sections / paragraphs: *What should be included? What is missing?*



1.	Introduction
2.	Experimental
2.1	Chemicals and reagents
2.2	Instruments
2.3	LC-MS/MS conditions
2.4	Preparation of standards and quality control samples
2.5	Sample preparation
2.6	Method validation
2.6.1	Linearity
2.6.2	Precision and accuracy
2.6.3	Selectivity
2.6.4	Recovery and ion suppression
2.6.5	Stability
2.7	Analysis of serum samples from intensive care patients
3.	Results and discussion
3.1	Method development
3.1.1	Sample pre-treatment
3.1.2	LC-MS/MS analysis
3.2	Method validation
3.2.1	Linearity
3.2.2	Accuracy and precision
3.2.3	Selectivity
3.2.4	Recovery and ion suppression
3.2.5	Stability
3.3.	Internal standard and matrix selection
3.4	Analysis of serum samples from intensive care patients
4.	Conclusion

Draft



Writing ≠ editing

- A draft is **not perfect**, do not try to make it perfect
- **Content:** general topics of the story you want to tell
- **Draft 1:** write the piece so YOU understand it
- **Draft 2:** Edit through the eyes of someone else. Be your reader.
 - Language (grammar, style, etc)
 - More elaboration on topics that seem obvious to you?
- **Draft 3:** Let someone else read it
 - Do they understand what is being said?
 - Structure and contents: logical structure, missing topics, etc?
 - Let someone else read it out loud: linguistic obstacles (incorrect grammar, sentences too long or complex, etc)

Goal:

Method:

Sample
preparation

Analytical
method

Method
validation

Application

Other
experiments

Results:

Extraction
optimization?

Method
optimization?

Validation
results?

Chromatogram?

Figures?

Tables?

Interesting
findings?

Application?

Conclusion:

Draft – exercise (1)

What's needed to achieve your project's main goal(s)?

Individually:

- Make a **sketch** of *how your poster would look like with the most ideal results/experiments* to show that you have accomplished **the main goal(s) of your project**.
- Base your poster on the knowledge you now have attained and ignore the results you have gained in the past weeks.
- Include also the *experiments you still need to do* to accomplish your final goal(s)

Group:

- Discuss the differences and similarities between the ideas.
- Try to reach a consensus on what to put in this poster?
- Does this change the course of experiments you have planned?



Draft – exercise (2)

What do you have and what needs to be done/added?

Group:

- Make a draft of a poster of the **results for your project so far**. *What would you present? Which data are relevant to reach your goals?*
- *Discuss within group: What data are missing, what data need confirmation, which experiments are the most informative?*

Between groups:

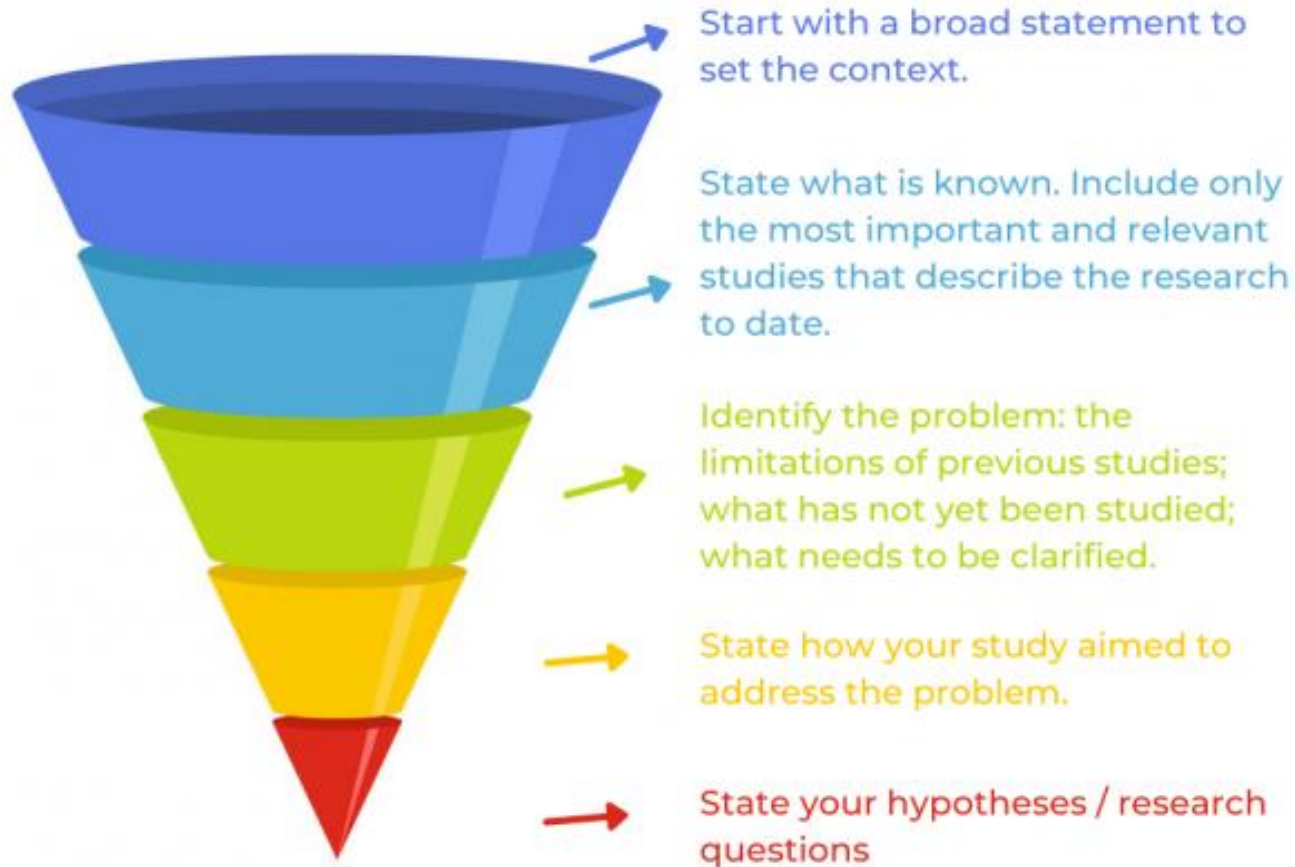
- Look at the poster of another group and discuss:

👍 **TOP:** What's clear?

💡 **TIP:** What needs more clarification / confirmation?



Introduction



Introduction



Bigger picture:

1. Set the context. Why is your research important?

Literature survey:

2. What is known?
3. What is unknown?

Your research:

4. How does your research fill these discrepancies?
5. What are your most important findings? Why should the reader read your paper?

Literature review



Group:

- Make an inventory:
 - What are the **topics** of your subject?
 - Have you found literature to substantiate your claims in the introduction?
 - **Reviews and meta-analysis papers:** Most information dense and cover broad topics → most useful to introduction
 - **Peer reviewed literature:** individual research
 - **Other sources:** can be used, but make sure they are verified by experts in the field (university websites, SOPs for industry, application notes, etc)
- What information is useful to your subject and what not?
- **Write a summary of this relevant background information**

Preparations next session:



Individual:

- Finalize the **outline** / structure of your report: add notes and ideas
- Describe and visualize at least **1 result**.

Group or individual:

- Write a summary of relevant background information for your project (**literature review**). This will be part of your introduction!
- Fill out the **Feedback form!**

Feedback session 1



<https://forms.gle/LvWjsToRhMYoYuS9A>

- How would you rate your own report writing skills on a scale from 1 (very bad) to 5 (very good)
- What difficulties do you experience in writing a report?
- What did you expect from the report writing seminar(s)?
- To what extent was your expectation met in session 1?
- List at least 3 things from session 1 that were helpful for you
- What could be improved in the **structure** (exercises, interaction, time management) of session 1?
- What could be improved in the **content** (topics) of session 1?
- What would you like to see covered in session 2?
- Any additional comments or feedback are welcome!